



**ASIA PACIFIC UNIVERSITY OF TECHNOLOGY AND INNOVATION
MOBILE & WEB MULTIMEDIA
INDIVIDUAL ASSIGNMENT**

Intake :
Lecturer :
Date Due :
Date Assigned :

LEARNING OUTCOMES

On completion of the module you should be able to demonstrate the following learning outcome:

- Construct appropriate mobile web multimedia application (C6, PL03)
- Report effective reasoning on the technologies utilized in the application (A3, PLO4)

PROJECT TITLES

You are required to produce an E-Commerce web application optimized for mobile devices.

Assume the role of a mobile web applications developer in an organization that specializes in designing and developing mobile-based solutions. The company has the opportunity to secure a contract with a major client, who has requested the design and development of a mobile web application that enables consumers to purchase products or services online.

Your task is to analyze the client's requirements and develop the mobile web application accordingly.

Your task in this assignment is to create an interactive and dynamic website using .NET technologies. You will need to meet various implementation requirements, but the actual content of the website is entirely up to you. The goal is to create a relatively small application that is extremely well designed, architected and implemented.

You are required to complete this assignment entirely by yourself. The website should consist of following **web pages** listed below.

1. Member Registration
 - a. Login
 - b. Manage Profile
2. Administrative Functions (Website Management)
 - a. Manage Users (RUD)
 - b. Manage Products (CRUD)
 - c. Manage Categories (CRUD)
 - d. Manage Orders
 - i. Pending
 - ii. Cancel
 - iii. Delivery
 - iv. Delivered
3. Product Catalog/ Product Listings
4. Shopping Cart
 - a. Current
 - b. History of purchases
5. Payment Support
 - a. Send email/Send notification
6. Product Delivery
 - a. Pending
 - b. Sent out for delivery
 - c. Delivered
7. Customer Feedback
 - a. Giving ratings to the product/transaction

The mobile website must demonstrate good usage and appropriateness of multimedia element. You are expected to provide a suitable interface by applying design principles and a fully functional navigation component.

In addition to the core requirements, you are encouraged to enhance your website by implementing **extra features** that can make your project more interactive, engaging, and modern. These may include :

- **Gamification elements** such as badges, reward points, loyalty programs and leaderboards
- **Personalization features** like wishlists, recently viewed products, product recommendations, and AI-based product recommendations
- **Social features** such as referral codes, user-generated content, social media logins and user-generated content
- **Customer retention tools** like abandoned cart notifications, discount coupons, flash sales, countdown timers, and live order tracking
- **Administrative enhancements** such as simple admin reports, sales dashboards with charts, stock alerts, and customer insights.

Before starting your assignment, you must inform the lecturer of your chosen project title.

REQUIREMENTS

You are required developed a detailed requirements specification for the website. The website must fulfill the basic requirements described below:

1. The focus of this assignment to develop a mobile web interactive and dynamic website
2. Your Web application should be able to be viewed in any standard Web browser, such as Microsoft Edge, Chrome, Mozilla or Opera. This means that you should produce only standard HTML and ECMA script for client-side processing.
3. Your Web application should store information in a server-side SQL Server database. Your Web application should both read and write this information.
4. The mobile Web based system application with user guide.
5. It is strongly recommended that you develop your application in an incremental fashion, where you first develop a working system that meets all of the minimum requirements before contemplating any of the optional requirements.

In addition, the website must fulfill the following criteria; i.e. it should:

1. Consist of 10 - 20 interlinked Web pages.
2. Demonstrate good usage of standard XHTML 1.0 or HTML5 elements.
3. Demonstrate good usage of standard Cascading Style Sheets (CSS 2.0 & 3.0).
4. Display appropriate and quality contents.
5. Demonstrate database connectivity & processing.
6. Using AttachDBFilename option for Microsoft SQL database. The database must be created in App_Data directory within the root directory of your website. The connection string must store in the website configuration file, web.config. This is to allow graders to test and run the website with zero or minimal settings.
7. Contain a registration page for new members.
8. Protect & restrict access to website resources using ASP.NET forms authentication and authorization.
9. Create and use ASP.NET Web forms user controls.
10. Make use of advanced Web controls such as Repeater, DataList, DetailView, GridView, and etc.
11. Use data binding to link Web controls to data sources.
12. Contain “member only” pages (require a valid login) to modify member profile, add to cart, view and update cart and perform relevant tasks which are permitted to members only.
13. Contain “admin only” pages (require a valid login) to maintain the database & perform administrative tasks.
14. Perform form validation.
15. Contain a feedback form for visitors to give their comments and suggestions.
16. Provide appropriate navigation support.
17. Perform proper file organization and naming convention.
18. Contain a static HTML page named ReadMe.html that displays the following:
 - A list of any (admin) user names or passwords that might be required by graders to test your Web application.
 - A summary to setup & configure your website
19. Run and test on a standalone Windows PC with Visual Studio and MS SQL Server Express.

20. Be viewable on multiple browsers and settings, and on any operating system. This means that you should produce only standard XHTML, CSS and JavaScript for client-side processing.

Note:

1. Your application should be accessible via a mobile browser regards of the type of mobile device.
2. The usage of Web page creation tools is permitted.
3. Verification for online payment is not required.
4. All other conditional checks that are necessary should be included.

DOCUMENTATION

The documentation should be in a standard report format with appropriate header and footer. There must be a cover page stating your name and ID. The documentation should contain at least 2000 words. The documentation needs to be submitted in Moodle. All other necessary project files to run the website should be submitted to Moodle as a zipped file.

Part 1: Proposal (Due Week 5)

- Approved E-Commerce Web Application
- Introduction
- Objectives
- Database Diagram (ERD)
- Wireframe – Mobile version
- Use Case Diagram
- Activity Diagram

Part 2: Report (Due Week 13)

The documentation should include the following:

1. Styles & Formats
 - Type face: Times New Roman/ Arial. Bold face, italic & lines can be used for emphasizing and to enhance readability
 - Font size: 12(except tittles and headings)
 - Margins: 1” from the left, right, top & bottom of the edges of the A4 paper
 - Spacing: maximum 1.5 lines between texts of a paragraph
 - Alignment: full justify
 - Headers and footers can be used
 - Each page must be numbered
 - APA Referencing should be followed.
2. Cover page, your cover page should contain the following items
 - Your name and ID
 - Intake code
 - Subject
 - Project title

- Date assigned (the date the report was handed out)
- Date completed (the date the report is due to be handed in)
- APU logo

3. Recommended Content

- Acknowledgement
- Table of Contents - The table of contents should have the topic title and reference page number attached to each topic.
- Introduction or Project Plan - In this section, briefly explain what your website is about and why you are creating it. Describe the goals of your project, who your target users are, and what the website will cover (scope). Include a simple timeline or schedule for development. Finally, list the main functions of your website and explain why you chose to add any extra features.
- Requirement Specification - In this section, write down what your website needs to do for its users (customers, admin, etc.). List all the main functions such as registration, shopping cart, payments, and order management. If you added extra features (like badges, wishlist, or coupons), include them here as part of your functional requirements.
- Design - In this section, show how your website is planned. This includes the database (tables and relationships), the layout of your pages, the navigational structure (how users move from one page to another), and the overall website architecture. Explain why you designed it this way. Add wireframes and simple UI mockups (drawings or screenshots) to show the look and flow of your pages.
- Implementation - In this section, explain how you built your website. Start by describing the design of your user interface (what the pages look like and how users interact with them). Next, explain how the system was developed and organized so it could work in a real-world situation. Finally, list the main steps you followed to create the system. If you added extra features (like live order tracking, dark mode, or multi-language support), describe how you coded them.
- Conclusions – In this section, look back at your project and explain what you have achieved. Check if your website met the goals you set in the beginning. Write about any problems or limitations you faced, and suggest how these could be fixed or improved in the future. Finally, explain how the extra features you added made the website better for users.
- References
- Appendices

4. User Manual

- A list of any (admin) user names or passwords that might be required by graders to test your web application.
- A summary to setup & configure your website.
- A simple walkthrough to use your website.

SUBMISSION

1. All documentation must be submitted online to Moodle.
2. All other necessary project files to run the website must be submitted to Moodle as a zipped file.

PRESENTATION

You are required to present your website to the lecturer, and the presentation will be judged accordingly.

ASSESSMENT CRITERIA

The assignment has a weight of 100%. Your effort will be assessed based on the following criteria.

No	Criteria	Mark Allocated	Score
Documentation (100)			
1	Project Proposal	10	
2	Introduction	20	
3	Requirement specification	20	
4	Design	20	
5	Implementation	20	
6	Conclusion, Document Styles and Formatting & User Guide	10	
			0
Implementation (90)			
6	Display Record - Dynamic content, Product catalogue, & etc. - Selected features	20	
7	Insert Record - registration, add new product, add to cart & etc. - Selected features	20	
8	Update & Delete Record - manage profile, products, shopping cart & etc. - Selected features	20	
9	User Authentication & Form Validation Selected features	20	
10	Mobile Web Layout & Appearance, Navigation design, content of the website and usability of the web pages	10	
			0
	Total Marks	190	0
	Percentage (%):	100%	0.00
Presentation (10/10)		Mark Allocated	Score
1	Presentation	10	

GRADING CRITERIA

Performance Indicators & Marking Guideline:

Grade	Indications	Score/10	Score/20
Distinction >70%	- Distinctive work will exhibit a high level of penetration in analysis and design, outstanding and clarity in communication, design and implementation. The supporting report should be exceptionally well-substantiated in its analysis, reasoning, and conclusions with each area. - In addition, at the advanced level the student will demonstrate an exceptional ability to synthesize innovative perspectives of the topic with work from other parts of the discipline.	9-10	16-20
Merit 65-74%	- Student should be able to show further evidence of knowledge, understanding the module material through analysis, design, prototype and implementation of the Web site. The supporting report should be well written and supported throughout in its analysis, reasoning and conclusions. - An advanced level will be indicated by student's reference to an integration of materials from previous programs of study.	7-8	12-15
Pass 50-64%	- The student will show the ability to apply appropriate techniques for analysis, design, prototype and implementation of the Web site. - There should be satisfactory evidence of knowledge, understanding the application material in the supporting report.	5-6	10-11
Marginal Fail 40-49%	Work at this level will generally be of low standard where it may even fail to meet at least 50% of the basic requirements listed above. Work at this level is not documented adequately, lacks referencing and does not display proper program structure would fall in this category. During demonstration of the project, the student is not able to explain briefly the work that has been done.	3-4	6-9
Fail 0-39%	Work at this level will generally be of very low standard where it may fail to meet at least 20% of the basic requirements listed above. Work at this level is not documented adequately, lacks referencing and does not display proper	0-2	0-5

	program structure would fall in this category. During demonstration, the student is not able to explain briefly the work that has been done.		
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